



COURSE NUMBER: CP1295

COURSE TITLE: Advanced JavaScript

COURSE DESCRIPTION:

This course is designed to provide students with an advanced understanding of JavaScript programming, including its principles, concepts, and best practices. The course will cover a range of topics including advanced language features, asynchronous programming, and front-end web development frameworks. Through lectures, hands-on exercises, and project work, students will learn how to write efficient and effective JavaScript code for a variety of applications.

PREREQUISITES: CP1210 - JavaScript

CO-REQUISITES: None

CARNEGIE UNIT: Two (2)

CREDIT VALUE: Three (3)

COURSE HOURS PER WEEK: Two (2)

LAB HOURS PER WEEK: Two (2)

REQUIRED TEXT:

Delameter, M., and Ruvalcaba, Z. (2020). *Murach's JavaScript and jQuery*. (4th ed.). Murach Publishing Fresno, CA. Print ISBN: 9781943872626, e-Text ISBN; 9781943872633

LEARNING RESOURCES:

<https://developer.mozilla.org/en-US/docs/Web/JavaScript>

MAJOR TOPICS:

- 1.0 JavaScript Objects, Functions, Events, and Data Structures
- 2.0 Creating, Extending, and Using Custom Objects
- 3.0 Tools, Objects, and Features of the Web Browser
- 4.0 Asynchronous JavaScript
- 5.0 JavaScript Modules

6.0 Working with JSON Data

LEARNING OBJECTIVES:

1.0 JavaScript Objects, Functions, Events, and Data Structures

- 1.1 Use the window and document objects
- 1.2 Manipulate the DOM while using the:
 - 1.2.1 Textbox objects
 - 1.2.2 Number objects
 - 1.2.3 Date objects
 - 1.2.4 String objects
- 1.3 Use of function expressions and function declarations
- 1.4 Attach function expressions and function declarations to events
- 1.5 Implement event handling in the browser
 - 1.5.1 Register event listeners that manipulate the DOM
 - 1.5.2 Register event listeners that validate data
- 1.6 Show proficiency in working with arrays through the effective application of the map, reduce, and filter functions

2.0 Creating, Extending, and Using Custom Objects

- 2.1 Create objects with customized methods and properties using object literals
- 2.2 Build custom objects by using constructors and the "new" keyword
- 2.3 Generate new objects by cloning object prototypes
- 2.4 Develop custom objects using the class syntax
- 2.5 Utilize objects in the development of basic web applications

3.0 Tools, Objects, and Features of the Web Browser

- 3.1 Use the web browser developer tools to identify errors
 - 3.1.1 Use the web browser developer tools to fix errors
- 3.2 Use of breakpoints to step through the execution of code
- 3.3 Add logging statements to trace the execution of code
- 3.4 Handle exceptions with try-catch statements
- 3.5 Demonstrate uses of cookies
- 3.6 Demonstrate uses of sessionStorage and localStorage from the Web Storage API
- 3.7 Utilize developer tools to manipulate cookies and web storage items by viewing, editing, and deleting them

4.0 Asynchronous JavaScript

- 4.1 Process asynchronous code in the JavaScript event loop
- 4.2 Create web applications using callbacks for writing asynchronous code
- 4.3 Create web applications utilizing the JavaScript Promise API for writing asynchronous code
- 4.4 Construct web applications using the async/await syntax for writing asynchronous code

5.0 JavaScript Modules

- 5.1 Implement the module design pattern (IIFE) to encapsulate data
- 5.2 Write applications using JavaScript modules
- 5.3 Describe the evolution of JavaScript modules using closures
 - 5.3.1 Create a modular JavaScript application using closures or the IIFE design pattern
- 5.4 Analyze how JavaScript modules are used in front-end libraries and frameworks to enhance code organization and maintainability
- 5.5 Create a modular front-end application using JavaScript modules in a library or framework, such as React or Angular

6.0 Working with JSON Data

- 6.1 Discuss the process of data serialization in web applications
 - 6.1.1 Discuss the process of data deserialization in web applications
- 6.2 Explain data formats including (but not limited to):
 - 6.2.1 csv
 - 6.2.2 xml
 - 6.2.3 binary
 - 6.2.4 JSON
- 6.3 Discuss the format of JSON data
 - 6.3.1 Use the global JSON object to serialize data with the stringify method
 - 6.3.2 Use the global JSON object to deserialize data with the parse method
- 6.4 Build a web application that uses JavaScript's Fetch API to retrieve JSON data from a public API

EVALUATION:

Assignments	30%
Tests (2)	20%
Project	20%
Final Exam	30%

DATE DEVELOPED: February 2023

DATE REVIEWED:

REVISION NUMBER: 1

DATE REVISED: November 2023

Effective PeopleSoft Date: September 2024

Note to instructor: Check PIRS to ensure this outline is the most current version.